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ITAI 1371

09/03/2026

Reflection Journal

Working out this lab gave our team a clearer perspective on the machine learning workflow and the various parts of the process that come together to yield results. We learned about the distinctions between supervised, unsupervised, and reinforcement learning as well as the relevance of knowing what type of data is being used. The Wine dataset helped us understand how features and classes can help to train a model and how preprocessing can influence the outcome of a model. Among the most helpful features of the lab was the ability to play with various features and compare outcomes. For instance, switching the features to flavanoids, color_intensity, and proline gave us 91.7% of accuracy while some other combination to alcohol, hue and proline turned out to yield 100%. These experiments illustrated the idea that switching features might help to achieve better results in model development.

Another important takeaway was learning how to evaluate a model beyond just looking at its accuracy score. Using a confusion matrix allowed us to see which classes were correctly predicted and where exactly mistakes were made. This had facilitated the interpretation of the results, showing that accuracy alone does not provide a full story. Another conclusion was that getting a higher accuracy doesn't necessarily mean that a model is better because outputs must be tested further and validated through some additional methods (e.g. cross validation). This part of the lab made clear that machine learning means getting results and understanding why the model functions in a particular way rather than just execute the code and accept its outputs.

The lab also played an important part in improving our technical skills. Some of us started becoming more familiar with Python, scikit-learn, and Matplotlib, while others were doing better on their machine learning libraries and visualizing the outcome of their work. In general, this experience proved to make machine learning less theoretical and more practical for the real world. Our key point is that creating an effective model is not just picking an algorithm. It is about analyzing the input data, picking up relevant features, preparing the data accurately, and training the model. Working through all those steps in a group had a great effect on our understanding of how to apply the methods we learned to real-life situations.